F79SP Stochastic Processes 2022/23
Solutions to Dec 2022 Exam Qns.

In lectures and tutorials, students have seen questions similar to all subquestions of the exam.

1. (a). This is a derivation of a coursework.
(ai) Indeed,

$$\begin{aligned}&\mathbf{P}(S_{n+1}=j \mid S_n=i, S_{n-1}=k_{n-1}, \ldots, S_1=k_1, S_0=a)\\ =&\mathbf{P}(\xi_{n+1}=j-i \mid S_n=i, S_{n-1}=k_{n-1}, \ldots, S_1=k_1, S_0=a).\end{aligned}$$

Since the conditioning events are determined by $\xi_1, \ldots, \xi_n$ that do not depend on ξ_{n+1}, we get that the probability above is $\mathbf{P}(\xi_{n+1}=j-i)$.
(**1 mark**)
This probability does not depend on n, this means the time-homogeneity.
(**1 mark**)
This probability depends only on the difference $j-i$, of the future and the current positions, this is the space-homogeneity.
(**1 mark**)

(aii) The set $\mathcal{Z}$ of all integers is a union of two disjont sets, Z_{odd} of odd numbers and Z_{even} of even numbers. Clearly, S_n is odd if and only if S_{n+1} is even. Therefore, the Markov chain is periodic with period $d \geq 2$. Next, d cannot be bigger than 2 since $p_{00}^{(2)} = \mathbf{P}(S_2 = 0 \mid S_0 = 0) = 2pq > 0$.
(**2 marks**)

(aiii) Stirling formula is $n! \sim \sqrt{2\pi n} n^n e^{-n}$, as $n \to \infty$. We know that the random walk returns to the initial state in $2n$ steps if it has n jumps up and n jumps down, i.e. n "successes" in $2n$ trials. The probability of that is

$$\frac{(2n)!}{n!n!}p^n q^n \sim \frac{\sqrt{4\pi n}(2n)^{2n}e^{-2n}}{2\pi n \cdot n^{2n}e^{-2n}}(pq)^n \sim \frac{1}{\sqrt{\pi n}}2^{2n}(pq)^n = \frac{(4pq)^n}{\sqrt{\pi n}}$$

(**2 marks**)
If $p \neq 1/2$, then $4pq < 1$ and the series $\sum_n (4pq)^n$ converges. If $p = q = 1/2$, then $4pq = 1$ and the series $\sum_n n^{-1/2}$ diverges.
(2 marks)

(bi) Due to space homogeneity, the first probability is equal to $\mathbf{P}(\text{there exists } n \text{ such that } S_n \leq 0 \mid S_0 = 10)$ that is the ruin probability $\min(1, (q/p)^{10})$. By the symmetry, the second probability is $\min(1, (p/q)^{10})$.

(**2 marks**)
If $p \le q$, then the first probability is 1, and if $p \ge q$, then the second probability is 1. Therefore, at least one of them is 1.
(**1 mark**)
Both are equal to 1 iff $p = q = 1$.
(**1 mark**)

(b2) Since $p > q$, the first probability in (bi) is less than 1 and the time to ruin is infinite with a positive probability. Therefore, $\mathbf{E}T_1 = \infty$.
(**1 mark**)
The use the symmetry to get that $\mathbf{E}T_2 = a/(1-2q)$ with $a = 10$. Thus, $\mathbf{E}T_2 = 30$.
(**2 marks**)
Total: **16 marks**.

2.

(a) One can draw a transition diagram that helps to see that there are two closed classes: $C_1 = \{1, 4, 5\}$ and $C_2 = \{6\}$ (absorbing). Indeed, $p_{11} + p_{14} + p_{15} = p_{51}p_{54} + p_{55} = p_{41} + p_{44} + p_{45} = 1$ and the states intercommunicate, $p_{14} > 0$, $p_{45} > 0$, $p_{51} > 0$. Therefore, C_1 is a class and a closed class. Further, C_2 is absorbing, $p_{66} = 1$.
(1 mark)

There are three states left, 2, 3 and 7. Neither can belong to a closed class since, e.g., $p_{74} > 0$ (and 7 is in an open class), then $p_{27} > 0$ (and 2 is in an open class) and $p_{31} > 0$ (and 3 is in an open class). Clearly, $p_{27} > 0$ and $p_{72} > 0$, so 7 and 2 intercommunicate and belong to the same class. Finally, $p_{i3} = 0$ for any $i \neq 3$, so 3 does not communicate with any other state. Thus, we have two open classes, $O_1 = \{2, 7\}$ and $O_2 = \{3\}$.
(1 mark)

(b) State 1 belongs to the closed class C_1. The class is finite and is aperiodic since, say, $p_{11} > 0$.
(1 mark)
Then there exists the limiting distribution that coincides with the only stationary distribution within this class. To find it, we have to solve a system

of four linear equations

$$\pi_1 p_{11} + \pi_4 p_{41} + \pi_5 p_{51} = \pi_1 \cdot 1/4 + \pi_4 \cdot 1/4 + \pi_5 \cdot 1/2 = \pi_1,$$
$$\pi_1 p_{14} + \pi_4 p_{44} + \pi_5 p_{54} = \pi_1 \cdot 1/4 + \pi_4 \cdot 1/4 + \pi_5 \cdot 1/4 = \pi_4,$$
$$\pi_1 p_{15} + \pi_4 p_{45} + \pi_5 p_{55} = \pi_1 \cdot 1/2 + \pi_4 \cdot 1/2 + \pi_5 \cdot 1/4 = \pi_5,$$

and

$$\pi_1 + \pi_4 + \pi_5 = 1.$$

(1 marks)

One of three first equations is redundant. Say, remove the last equation and solve the remaining system. From the second and the very last equations we get immediately that $\pi_4 = 1/4$, then $\pi_1 + \pi_5 = 3/4$ and $\pi_5 = 3/4 - \pi_1$. The answer is: $(\pi_1, \pi_4, \pi_5) = (7/20, 1/4, 8/20)$. Thus, the limiting distribution is $\pi^{(\mathbf{1})} = (7/20, 0, 0, 1/4, 2/5, 0, 0)$.

(2 marks)

(c) It remains to find stationary distribution within class C_2 and then provide a general form of the stationary distribution. The stationary distribution in class C_2 is obviously $\pi^{(\mathbf{2})} = (0, 0, 0, 0, 0, 1, 0)$. **(1 mark)**

Then the general form of a stationary distribution is
$\gamma_1 \pi^{(\mathbf{1})} + \gamma_2 \pi^{(\mathbf{2})} = (7\gamma_1/20, 0, 0, \gamma_1/4, 8\gamma_1/20, \gamma_2, 0)$
where $\gamma_1, \gamma_2 \geq 0$ and $\gamma_1 + \gamma_2 = 1$.

(2 marks)

(d) We have to find absorption probabilities

$$\alpha_i^{(j)} = \mathbf{P}(\text{Markov chain is absorbed in class } C_j \mid X_0 = i).$$

Since the number of states is finite and there are only two classes, $\alpha_i^{(1)} + \alpha_i^{(2)} = 1$, for any i. Write $\alpha_i = \alpha_i^{(1)}$, for short. Clearly, $\alpha_1 = \alpha_4 = \alpha_5 = 1$ and $\alpha_6 = 0$. (**2 marks**)

Further, use the formula:

$$\alpha_i = \sum_j p_{ij} \alpha_j.$$

Then we get immediately that $\alpha_2 = \alpha_7$ and

$$\alpha_7 = 1/4 \cdot \alpha_2 + 1/4 \cdot 1 + 1/4 \cdot 0 + 1/4 \cdot \alpha_7.$$

Therefore, $\alpha_2 = \alpha_7 = 1/2$.

Further,

$$\alpha_3 = 1/3 \cdot 1 + 1/3 \cdot \alpha_2 + 1/3 \cdot \alpha_3$$

and $\alpha_3 = \alpha_3^{(1)} = 3/4$ and, therefore, $\alpha_3^{(2)} = 1/4$.

(2 marks)

So the limiting distribution is

$\alpha_3^{(1)}\pi^{(\mathbf{1})} + \alpha_3^{(2)}\pi^{(\mathbf{2})} = (21/80, 0, 0, 3/16, 3/10, 1/4, 0)$.

(1 mark)

(e) Let $f_{ij}^{n)} = \mathbf{P}(X_n = j, X_{n-1} \neq j, \ldots, X_1 \neq j \mid X_0 = i)$ be the probability of the first visit to state j at time n if the Markov chain starts from i. Then we have to find the sum $\sum_n f_{36}^{(n)}$.

(1 mark)

Clearly, we can get to 7 via 2 only. Then $f_{36}^{(1)} = 0$, $f_{36}^{(2)} = p_{32}p_{27} = 1/3$ and, for $n = 3, 4, \ldots$, we have $f_{36}^{(n)} = p_{33}^{(n-2)}p_{32}p_{27} = (1/3)^{n-1}$. Thus, the target probability is

$$1/3 + (1/3)^2 + \ldots + (1/3)^k + \ldots = \frac{1/3}{1 - 1/3} = \frac{1}{2}.$$

(2 marks)

(f) Starting from 2, the Markov chain jumps to 7 and then arrives to one of two closed classes after several visits of 2 and 7. Denote by T_i the time to arrive to any closed class starting from state i at time 0, and by T_i' the time to arrive to any closed class starting from state i at time 1. Then

$$T_2 = 1 + T_7'$$

and

$$T_7 = 1 + \mathbf{I}(B_2)T_2' + \mathbf{I}(B_7)T_7'$$

where B_2 means "jump from 7 to 2" and B_7 means "jump from 7 to 7" at the first step.

(2 marks)

Then

$$\mathbf{E}T_2 = 1 + \mathbf{E}T_7' = 1 + \mathbf{E}T_7 \quad \text{(due to time-homogeneity)}$$

and

$$\mathbf{E}T_7 = 1 + 1/4\mathbf{E}T_2 + 1/4\mathbf{E}T_7 \quad \text{(again, by time-homogeneity)}.$$

Thus, $\mathbf{E}T_7 = 5/2$ and $\mathbf{E}T_2 = 7/2$.
(2 marks)

(g) We have

$$\mathbf{P}(X_0 = 2 \mid X_2 = 4) = \frac{\mathbf{P}(X_0 = 2, X_2 = 4)}{\mathbf{P}(X_0 = 2, X_2 = 4) + \mathbf{P}(X_0 = 3, X_2 = 4) + \mathbf{P}(X_0 = 4, X_2 = 4)}.$$

(1 mark)

Clearly, the numerator is $1/3 \cdot p_{27}p_{74} = 1/3 \cdot 1/4$. Further, $\mathbf{P}(X_0 = 3, X_2 = 4) = 1/3 \cdot p_{31}p_{14} = 1/3 \cdot 1/12$ and

$$\mathbf{P}(X_0 = 4, X_2 = 4) = 1/3 \cdot (p_{41}p_{14} + p_{44}p_{44} + p_{45}p_{54}) = 1/3 \cdot 1/4.$$

Thus, the target probability is

$$\frac{1/4}{1/4 + 1/12 + 1/4} = 3/7.$$

(2 marks)
(Total 24 marks)

3.

(a) Clearly,

$$p_{11} = p_1 = 1/4,\ p_{12} = p_2 = 1/4.\ p_{13} = p_3 = 1/4.$$

Further,

$$\begin{aligned}
p_{21} &= 2p_0p_1 = 2/16 = 1/8;\\
p_{22} &= 2p_0p_2 + p_1p_1 = 1/8 + 1/16 = 3/16;\\
p_{23} &= 2p_0p_3 + 2p_1p_2 = 1/8 + 1/8 = 4/16 = 1/4;\\
p_{31} &= 3p_1p_0^2 = 3/64;\\
p_{32} &= 3p_2p_2^2 + 3p_1^2p_0 = 6/64 = 3/32;\\
p_{33} &= 3p_3p_0^2 + 6p_2p_1p_0 + p_1^3 = 10/64 = 5/32.
\end{aligned}$$

(4 marks)

(b)

$$\begin{aligned}
\mathbf{P}(\tau_1(S) \geq 3) &= p_1(p_1 + p_2 + p_3) + p_2(p_{21} + p_{22} + p_{23}) + p_3(p_{31} + p_{32} + p_{33})\\
&= \frac{1}{4} \cdot \frac{3}{4} + \frac{1}{4} \cdot \frac{9}{16} + \frac{1}{4} \cdot \frac{19}{64} = \frac{103}{256}.
\end{aligned}$$

(2 marks)

(c) For $i = 1, 2, 3$, we get the system of three equations:

$$\mathbf{E}\tau_i(S) = 1 + \sum_{j=1}^{3} p_{ij}\mathbf{E}\tau_j(S),$$

where the probabilities p_{ij} are given in solution to question 3(a) above.
(2 marks)

4. Questions (a) and (b) are coursework.

(ai) Let $\psi = \psi_1$ and $\alpha = \alpha_1$.

$$\begin{aligned}\mathbf{P}(\psi > t+u \mid \psi > t) &= \frac{\mathbf{P}(\psi > t+u, \psi > t)}{\mathbf{P}(\psi > t)} \\ &= \frac{\mathbf{P}(\psi > t+u)}{\mathbf{P}(\psi > t)} \quad \text{since } \mathbf{P}(AB) = \mathbf{P}(A) \text{ if } B \subseteq A \\ &= \frac{e^{-\alpha(t+u)}}{e^{-\alpha t}} = e^{-\alpha u}.\end{aligned}$$

(1 mark)

(aii) This is Exponential distribution with parameter $\alpha_1+\alpha_2+\alpha_3$. Indeed, for any $t > 0$,

$$\begin{aligned}\mathbf{P}(\min(\psi_1, \psi_2, \psi_3) > t) &= \mathbf{P}(\psi_1 > t, \psi_2 > t, \psi_3 > t) \\ &= \mathbf{P}(\psi_1 > t)\mathbf{P}(\psi_2 > t)\mathbf{P}(\psi_3 > t) \quad \text{by independence} \\ &= e^{-\alpha_1 t}e^{-\alpha_2 t}e^{-\alpha_3 t} = e^{-(\alpha_1+\alpha_2+\alpha_3)t}.\end{aligned}$$

(2 marks)

(bi) The intensity λ is the sum of intensities $\lambda_1 + \lambda_2 + \lambda_3$.
(**1 mark**) We need to show that, in the superposed process $N(t)$, the number of points $N(t, t+u)$ in any time interval $(t, t+u)$ is Poisson with parameter λu, and that the numbers of points in any number of disjoint intervals are mutually independent. The first result follow since

$$N(t, t+u) = N_1(t, t+u) + N_2(t, t+u) + N_3(t, t+u)$$

where $N_i(t, t+u)$ is Poisson with parameter $\lambda_i u$, for $i = 1, 2, 3$ and since there three summands are mutually independent.

(**2 marks**)
The second result follows from the fact that functions of mutually independent random variables are mutually independent, too. Say, for two disjoint intervals $(t_1, t_1 + u_1)$ and $(t_2, t_2 + u_2)$, the random vectors $(N_1(t_1, t_1 + u_1), N_2(t_1, t_1+u_1), N_3(t_1, t_1+u_1))$ and $(N_1(t_2, t_2+u_2), N_2(t_2, t_2+u_2), N_3(t_2, t_2+u_2))$ are mutually independent. Then, the sums of their coordinates are mutually independent too.
(**1 mark**)

(bii) Since first points of Poisson processes have Exponential distributions, the distribution of the first point has Exponential distribution with parameter $\lambda_1 + \lambda_2 + \lambda_2$, as follows from (a2).
(2 marks)

(biii) The distribution of the second point is Gamma with parameters $(2, \lambda)$. This follows from the properties of the Poisson process.
(**2 marks**)

(ci) There are three options: either $N(4, 6) = 2$ and $N(6, 10) = 0$ or $N(4, 6) = 1$ and $N(6, 10) = 1$ or $N(4, 6) = 0$ and $N(6, 10) = 2$. We have

$$\mathbf{P}(N(6,12) = 4 \mid N(4,10) = 2) = \frac{\mathbf{P}(N(6,12) = 4, N(4,10) = 2)}{\mathbf{P}(N(4,10) = 2)}$$
$$= \frac{\mathbf{P}(N(4,6) = 2, N(6,10) = 0, N(10,12) = 4) + \mathbf{P}(N(4,6) = 1, N(6,10) = 1, N(10,12) = 3)}{\mathbf{P}(N(4,10) = 2)}$$
$$+ \frac{\mathbf{P}(N(4,6) = 0, N(6,10) = 2, N(10,12) = 2)}{\mathbf{P}(N(4,10) = 2)}.$$

(**2 marks**)

In the numerator, the probabilities of intersection of events are the products of probabilities, by independence. We know that $N(4, 6)$ has Poisson distribution with parameter $\lambda_1 = 2\times 2 = 4$, $N(6, 10)$ has Poisson with parameter $\lambda_2 = 8 \times 4 = 32$, $N(10, 12)$ has Poisson with parameter $\lambda_3 = 8 \times 2 = 16$, and $N(4, 10)$ has Poisson with parameter $\lambda_4 = 36$. then the target probability is

$$\frac{\frac{\lambda_1^2}{2!}\frac{\lambda_2^0}{0!}\frac{\lambda_3^4}{4!} + \frac{\lambda_1^1}{1!}\frac{\lambda_2^1}{1!}\frac{\lambda_3^3}{3!} + \frac{\lambda_1^0}{0!}\frac{\lambda_2^2}{2!}\frac{\lambda_3^2}{2!}}{\lambda_4^2} 2! \cdot e^{-\lambda_1-\lambda_2-\lambda_3+\lambda_4}.$$

Students are aware that the solution may be stopped here, without giving a final answer.

(3 marks)

(cii) The total intensity per day is $2 \times 6 + 8 \times 18 = 156$. The mean of individual claim size is $3 \times 2/3 + 6 \times 1/3 = 4$. Then, by the corresponding theorem from the course, the total expected value is $156 \times 4 = 624$.

(2 marks)

We know another theoretical result, that the conditional distribution is Binomial with parameters 5 and 1/3. Then the probability to have at most 2 "successes" is

$$\frac{5!}{5!0!}\left(\frac{1}{3}\right)^0\left(\frac{2}{3}\right)^5 + \frac{5!}{4!1!}\left(\frac{1}{3}\right)^1\left(\frac{2}{3}\right)^4 + \frac{5!}{3!2!}\left(\frac{1}{3}\right)^2\left(\frac{2}{3}\right)^3.$$

This is computable and equal to $\frac{192}{243} = \frac{64}{81}$.

(3 marks)

5. (a) Here is it:

$$P = \begin{pmatrix} 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 1/3 & 1/3 & 0 & 1/3 & 0 \end{pmatrix}$$

(2 marks)

(b) Since $p_{14} = p_{41} = 1$, $C_1 = \{1, 4\}$ is a closed class. Since $p_{22} = 1$, $C_2 = \{2\}$ is a closed (absorbing) class. Since $p_{51} > 0$, state 5 must belong to an open class. Since $p_{35} > 0$, state 3 must also belong to an open class. Since $p_{35} > 0$ and $p_{i3} = 0$, for all i, state 3 and 5 do not intercommunicate and, therefore, belong to different open classes, $O_1 = \{3\}$ and $O_5 = \{5\}$.

(2 marks)

(c) The answer is "no": for the "yes" answer, within each closed class, the departure rates from all states must be the same. It is fine with class C_2 (because it contains only one state). However, $q_{11} = -3 \neq q_{44} = -4$ in class C_1.

(2 marks)

(d) For a finite time-homogeneous Markov process, the limiting distribution exists for all initial values.

In a finite Markov chain, there is the limiting distribution for any initial state if the Markov chain is aperiodic in each closed class. The absorbing class is

aperiodic, so if $X_0 = 2$, the limiting distribution exists and is the same as the stationary $(0, 1, 0, 0, 0)$. However, the class C_1 is periodic with period 2. Therefore, for $X_0 = 1$ and $X_0 = 4$, the limiting distributions for the Markov chain do not exist.
(3 marks)
Now we show that the limiting distributions do exist if $X_0 = 5$ or $X_0 = 3$.
If $X_0 = 5$, then probabilities to enter 1 or 4 are equal. Therefore, given that at some time the Markov chain enters class C_1, its distribution at the entrance time is uniform on C_1, which is the stationary distribution. The same reasoning applies to $X_0 = 3$ because then the Markov chain immediately goes to $X_1 = 3$. **(2 marks)**